1495 UNIVERSITY OF ABERDEEN | CELEBRATING 525 YEARS 1495 – 2020

ABERDEEN 2040

Revision – Week 4

Data Mining & Visualisation

Lecture 31 & 32

2024

Time Series

Example Question

Question 2 from **cs4031w2010-2011.pdf**

(a) What is PAA (Piecewise Aggregate Approximation) representation of a time series?

PAA is a technique used to reduce the dimensionality of time series data, allowing us to approximate and simplify the time series data while maintaining its characteristics.

Example Question

(b) Consider the following time series data of goals scored by Diego Maradona and Pele playing for their respective national teams. The first column shows the years and the second column shows the corresponding goals for Maradona. The third column shows the data from the second column (goals data) in the normalized (scaled) form. The fourth column shows Pele's goals while the fifth column shows Pele's goals in the normalized form.

Year	Maradona Goals	Maradona Goals Normalized	Pele Goals	Pele Goals Normalized
1	0	-0.70	2	-0.83
2	0	-0.70	9	1.06
3	3	-0.08	11	1.60
4	7	0.73	4	-0.29
5	1	-0.49	0	-1.37
6	2	-0.29	8	0.79
7	0	-0.70	7	0.52
8	0	-0.70	2	-0.83
9	6	0.53	9	1.06
10	17	2.77	5	-0.02
11	4	0.12	0	-1.37
12	1	-0.49	4	-0.29

Example Question

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i) Compute a 3 segment PAA representations of the goals time series data (i.e. the number of segments in the PAA equal to 3) for each player showing all the major steps.

We know that we want 3 segments.

Therefore, each segment will have 4 years (12 / 3 = 4).

Segment 1: Years 1—4 Maradona goals: {-0.70, -0.70, -0.08, 0.73}

Segment 2: Years 5—8 Maradona goals: {-0.49, -0.29, -0.70, -0.70}

Segment 3: Years 9—12 Maradona goals: {0.53, 2.77, 0.12, -0.49}

Example Question

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Segment 1: Maradona goals: {-0.70, -0.70, -0.08, 0.73} mean = 0.65 / 4 = 0.16

Segment 2: Maradona goals: {-0.49, -0.29, -0.70, -0.70} mean = -2.18 / 4 = 0.55

Segment 3: Maradona goals: {0.53, 2.77, 0.12, -0.49} mean = 2.93 / 4 = 0.73

Maradona PAA: {0.16, 0.55, 0.73}

Example Question

i) Compute a 3 segment PAA representations of the goals time series data (i.e. the number of segments in the PAA equal to 3) for each player showing all the major steps.

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We know that we want 3 segments.

Therefore, each segment will have 4 years (12 / 3 = 4).

Segment 1: Pele goals: {-0.83, 1.06, 1.60, -0.29} mean = 1.54 / 4 = 0.39

Segment 2: Pele goals: {-1.37, 0.79, 0.52, -0.83} mean = -0.89 / 4 = -0.22

Segment 3: Pele goals: {1.06, -0.02, -1.37, -0.29} mean = -0.62 / 4 = 0.16

Pele PAA: {0.39, -0.22, 0.16}

Example Question

(ii) What is SAX (Symbolic Aggregate Approximation) representation of a time series? Write two advantages of SAX representation over other structural representations of time series.

SAX is a is a technique used to further compress time series data obtained from PAA by transforming numerical values into symbols.

Advantages:

Dimensionality Reduction. SAX enables the representation of long time series with a compact symbol sequence.

Pattern Recognition. SAX captures important patterns and trends in the data.

Interpretability: SAX allows for easy interpretation and comparison of time series.

Example Question

(iii) For the two PAAs from (b) (i) compute the SAX representations using the following break point data.

Alphabet	Breakpoint 1	Breakpoint 2
a	Negative Infinity	< -0.67
b	>= -0.67	< 0
c	>= 0	< 0.67
d	>= 0.67	Positive Infinity

Example Question

Alphabet	Breakpoint 1	Breakpoint 2
a	Negative Infinity	< -0.67
b	>= -0.67	< 0
c	>= 0	< 0.67
d	>= 0.67	Positive Infinity

Recall our previous PAA values.

Simply map these to the corresponding SAX values.

Maradona PAA: {0.16, 0.55, 0.73}

Maradona SAX: { c, c, d }

Pele PAA: {0.39, -0.22, 0.16}

Pele SAX: { c, b, c }

Example Question

(c) In the context of similarity matching of time series, briefly describe the term ‘lower bounding distance’ and state its importance for time series data mining.

Within the context of similarity matching of time series (such as through Dynamic Time Warping), lower bounding distance is a technique used to reduce the computational load in calculations.

It involves calculating a simpler, quicker-to-compute distance that serves as a lower bound to the true distance between sequences.

This allows us to efficiently filter out sequence pairs that are unlikely to be similar, thus avoiding computationally intensive calculations on all data pairs.

Example Question

(d) Consider the original time series and their SAX representations computed above.

i. Inspect the SAX representations from (b).(iii) above and comment on the similarity between the performance of the two players over the given time span.

ii. Inspect the original times series for each player in the table above and comment on the similarity between the performance of the two players over the given time span.

iii. Based on the comments from (d).(i) and (d).(ii) above, and the notion of lower bounding distance from 2.(c) above, justify the use of SAX representations for time series similarity matching.

Example Question

(d) Consider the original time series and their SAX representations computed above.

i. Inspect the SAX representations from (b). (iii) above and comment on the similarity between the performance of the two players over the given time span.

Maradona SAX: { c, c, d }

Pele SAX: { c, b, c }

We can see from the SAX representations that while Maradona and Pele performed comparatively similar over years 1—4, Maradona appeared to outperform Pele in years 5—8 and 9—12 in terms of goals scored (normalized).

Example Question

(d) Consider the original time series and their SAX representations computed above.

ii. Inspect the original times series for each player in the table above and comment on the similarity between the performance of the two players over the given time span.

Within the original data, despite having all of the information at hand, it is more difficult to assess each player's comparative performance over the duration of the time span.

Year	Maradona Goals	Maradona Goals Normalized	Pele Goals	Pele Goals Normalized
1	0	-0.70	2	-0.83
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Example Question

(d) Consider the original time series and their SAX representations computed above.

iii. Based on the comments from (d).(i) and (d).(ii) above, and the notion of lower bounding distance* from 2.(c) above, justify the use of SAX representations for time series similarity matching.

SAX allows us to simplify the raw data through dimensionality reduction, and to compare each player's performance over a given time span (4 years). This is comparatively easier than working with the raw data, where variability across years is not accounted for.

* Note, we haven't discussed lower bounding distance with regard to SAX, **<u>and this will not be in the exam</u>**, so we will ignore this part. But essentially the idea of reducing computational complexity by pruning unnecessary calculations is the same.

Decision Trees

Levels of Measurement

Levels of Measurement

Statistically, we define four levels of measurement for data: Nominal, Ordinal, Interval, and Ratio. Consider the following attributes, and point out which levels they belong to:

i. Country of origin
ii. Temperature in Celsius
iii. Temperature in Kelvin
iv. Exam grade: {A; B; C; D; F}
v. Age
vi. Quality of food: {bad; neither bad nor good; good}

Levels of Measurement

Statistically, we define four levels of measurement for data: Nominal, Ordinal, Interval, and Ratio. Consider the following attributes, and point out which levels they belong to:

(Nominal) i. Country of origin

(Interval) ii. Temperature in Celsius

(Ratio) iii. Temperature in Kelvin

(Ordinal) iv. Exam grade: {A; B; C; D; F}

(Ratio) v. Age

(Ordinal) vi. Quality of food: {bad; neither bad nor good; good}

Descriptive Statistics

Descriptive Statistics

Consider the following samples:

1	3	2	1	2	3

Calculate the **mean**, **standard deviation**, and **median** for the above samples.

Then calculate the **z-score** for **each** of the above samples.

Descriptive Statistics – Mean

Consider the following samples:

1	3	2	1	2	3

$\bar{x} = 2$

Recall that for samples:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{(1+3+2+1+2+3)}{6} = \frac{12}{6} = 2$$

Descriptive Statistics – Standard Deviation

Consider the following samples:

1	3	2	1	2	3

$$\bar{x} = 2$$
$$s = 0.89$$

Recall that for samples:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{N} (x_i - \bar{x})^2}$$

$$s = \sqrt{\frac{(1-2)^2+(3-2)^2+(2-2)^2+(1-2)^2+(2-2)^2+(3-2)^2}{6-1}}$$

$$s = \sqrt{\frac{1+1+0+1+0+1}{5}} = \sqrt{\frac{4}{5}} = \sqrt{0.8} = 0.89$$

Note

In the exam, you will not need a calculator.

Any advanced calculations you need to perform will be provided to you.

Descriptive Statistics – Median

(reordered)

Consider the following samples:

1	1	2	2	3	3

$$\bar{x} = 2$$
$$s = 0.89$$
$$\mathrm{med}(x) = 2$$

Recall:

$$\text{If } n \text{ is even, } \mathrm{med}(x) = \frac{x_{(n/2)} + x_{((n/2)+1)}}{2}$$

$$\mathrm{med}(x) = \frac{x_{(6/2)} + x_{((6/2)+1)}}{2} = \frac{x_{(3)} + x_{(4)}}{2} = \frac{2+2}{2} = \frac{4}{2} = 2$$

Descriptive Statistics – z-score

Consider the following samples:

1	3	2	1	2	3

$$\bar{x} = 2$$
$$s = 0.89$$
$$\text{med}(x) = 2$$

Recall:

$$z = \frac{(x - \bar{x})}{s} = \frac{(1-2)}{0.89} = \frac{-1}{0.89} = -1.12$$
$$= \frac{(2-2)}{0.89} = \frac{0}{0.89} = 0$$
$$= \frac{(3-2)}{0.89} = \frac{1}{0.89} = 1.12$$

Relationships in Data

Relationships in Data

- Briefly explain what you understand by the term ‘a weak positive correlation’.
- Briefly explain what you understand by the term ‘a strong negative correlation’.
- Briefly explain what you understand by the term ‘no correlation’.
- Briefly explain what you understand by the term ‘correlation does not imply causation’.

Relationships in Data

- Briefly explain what you understand by the term ‘a weak positive correlation’.

Correlation is a measurement of the strength of relationship between two variables.

A weak positive correlation would have a relatively small positive correlation coefficient. In other words, as variable x increases, variable y will have a slight tendency to increase.

Relationships in Data

- Briefly explain what you understand by the term ‘a strong negative correlation’.

A strong negative correlation would have a relatively large inverse correlation coefficient, representing two variables which are tightly and inversely associated with each other. In other words, as variable x increases, variable y will have a strong tendency to decrease.

Relationships in Data

- Briefly explain what you understand by the term ‘no correlation’.

No correlation would refer to a correlation coefficient close to 0. In other words, there is little to no relation between variables x and y.

Relationships in Data

- Briefly explain what you understand by the term ‘correlation does not imply causation’.

The term ‘correlation does not imply causation’ refers to the fact that correlation and causation are distinct terms. Correlation is a measure of the strength of an association between two variables, whereas causation is the process of one event causing or producing another event. From a given correlation coefficient, we cannot know whether causation exists or not.